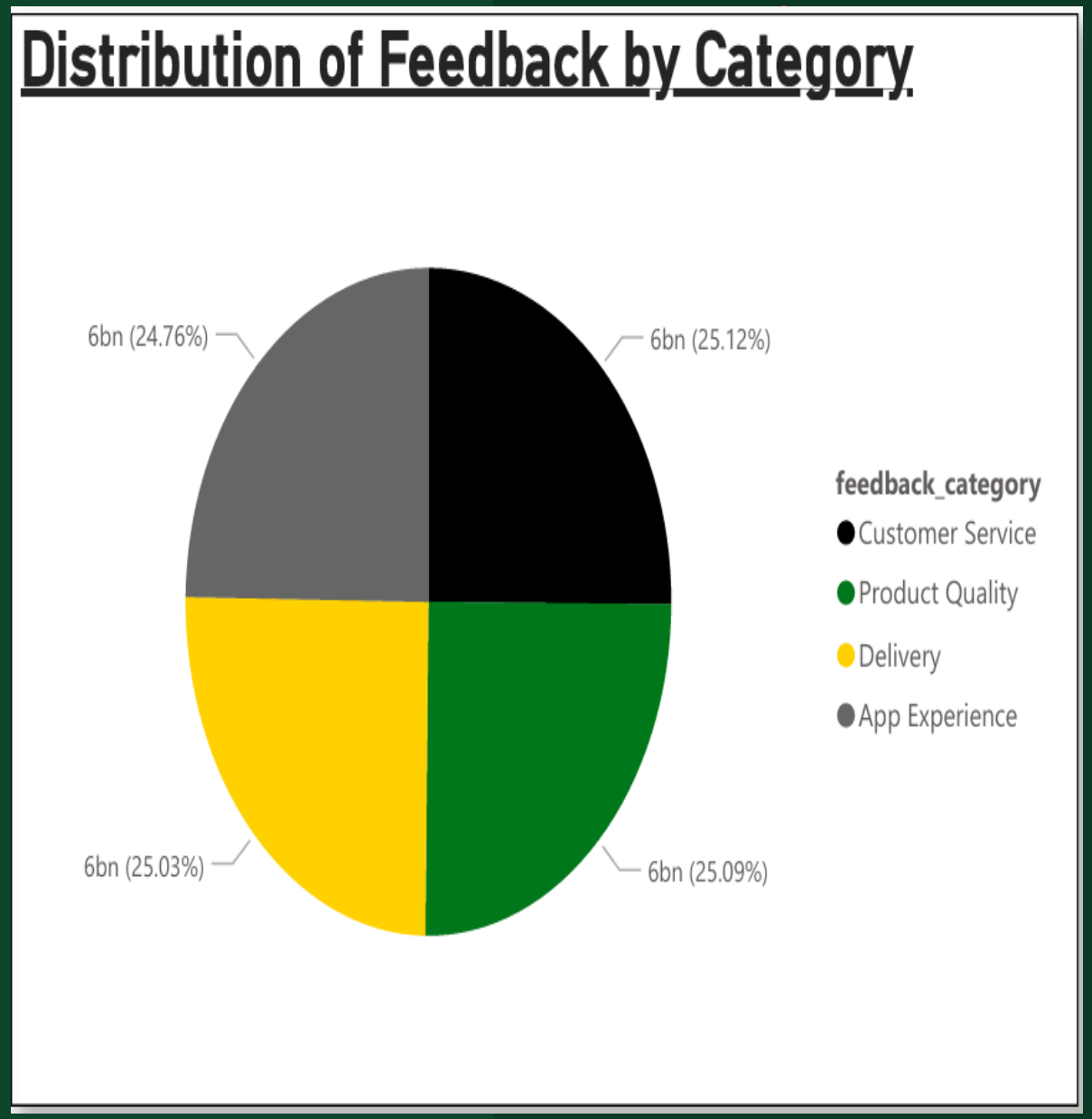
Blinkit Sales Analysis:-

How shall we import data from all 11 tables into Power BI?

In Power BI Desktop, data import is done using the "Get Data" option from the Home ribbon. To import multiple tables, you first click on "Get Data" and select the appropriate data source type (such as Excel, SQL Server, CSV, or other formats depending on where your data is stored). After connecting to the data source, you can preview and select multiple tables you want to load into Power BI. Once selected, you click on "Load" to bring these tables into Power BI's data model. After loading, all 11 tables will appear in the Fields pane, ready for further transformation, analysis, or visualization. If any data cleaning or transformation is needed before loading, you can use the Power Query Editor to shape the data as required before it enters your model.

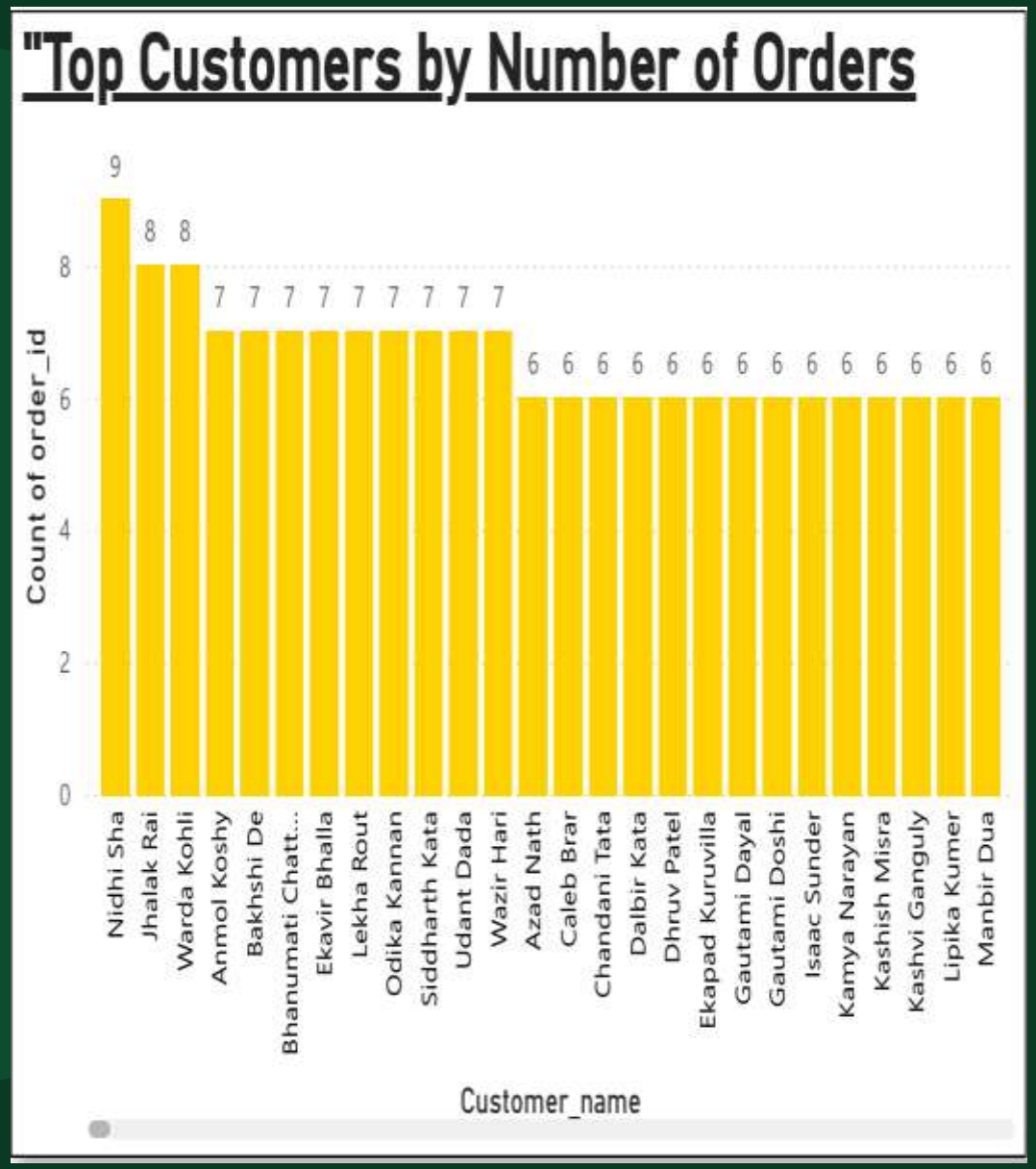
How do we create relationships between the tables in Power BI?

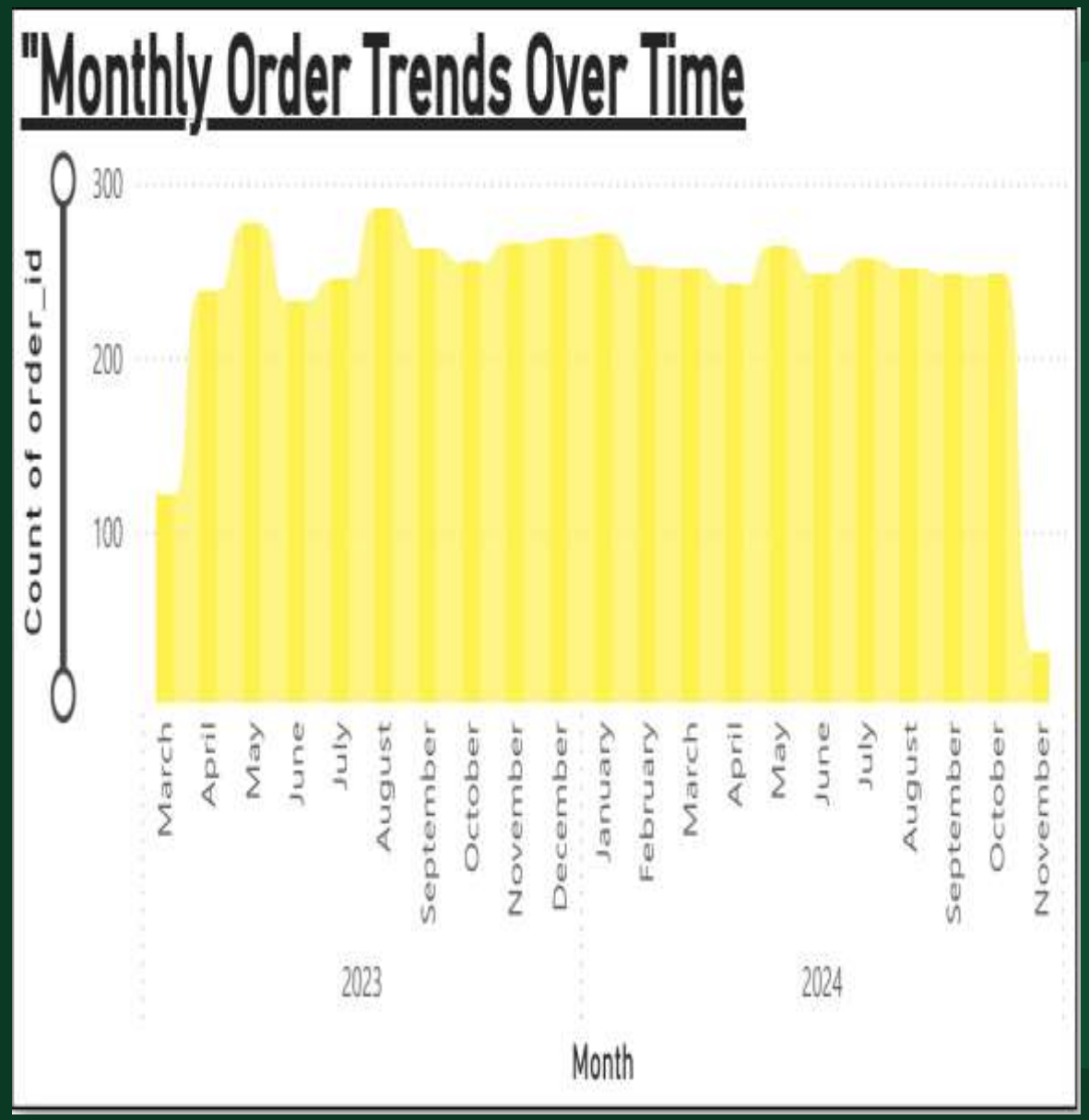
After importing multiple tables into Power BI, you need to establish relationships between them to build a unified data model. Relationships are created by matching common fields (keys) between tables — for example, Customer ID between the Customer table and the Orders table. To create a relationship, go to the "Model" view where all tables are shown as boxes. Drag and drop the related field from one table to the corresponding field in the other table. Power BI will automatically suggest the relationship type (typically one-to-many or many-to-one). You can also click on "Manage Relationships" from the Home ribbon to manually create or edit relationships, define cardinality, and configure cross-filter direction. Correctly defining relationships allows Power BI to join tables and calculate measures accurately across multiple tables when creating reports and visuals.



I imported the customer feedback data into Power BI and created a pie chart to show the different categories of feedback received. The chart displays four categories: Customer Service, Product Quality, Delivery, and App Experience. Each segment of the pie represents the percentage of total feedback that belongs to each category. This visualization helps identify which area of the business receives more feedback from customers, indicating which aspects of service or product experience require more attention or improvement.

Using the order data, I created a bar chart that displays the top customers based on the number of orders they have placed. Each bar represents a customer, and the height of the bar indicates how many orders that customer has made. This visualization helps easily identify the most active and loyal customers who contribute the most to overall sales, allowing the business to target these customers for loyalty programs or special offers.





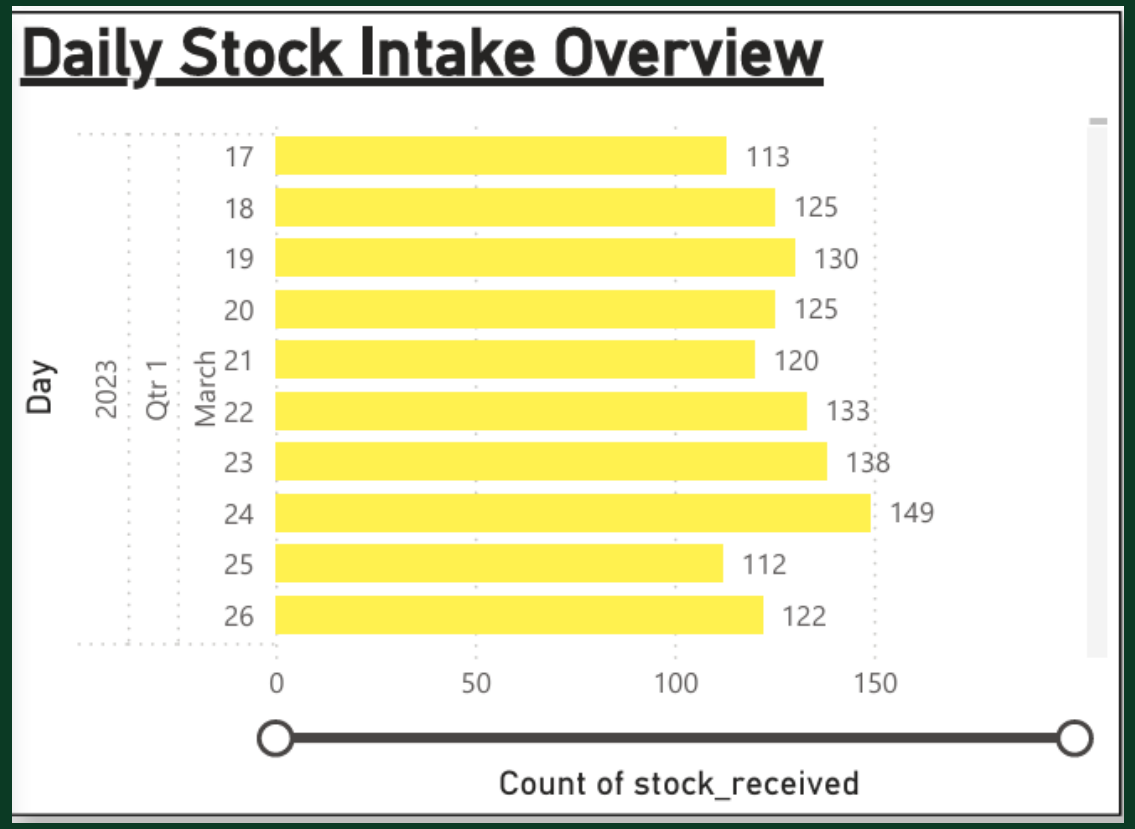
I organized the order data by month and used a line chart to show how the number of orders changed over time. The horizontal axis shows the months, and the vertical axis shows the total number of orders placed in each month. This visual helps identify seasonal trends and fluctuations in customer demand, allowing the business to plan inventory and marketing activities accordingly during peak and low seasons.

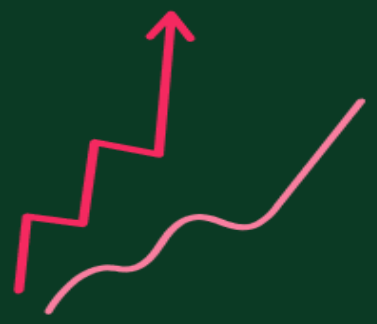
Customer Directory with Contact Information

Customer id	Customer name	Email id
80315623	Varenya Banik	aachalsarma@example.cor
70230254	Hamsini Sandhu	aachalsheth@example.net
14161586	Aachal Nazareth	aadhya71@example.org
61226563	Chameli Kala	aadhya91@example.com
74735607	Naksh Amble	aadhyabhalla@example.co
71615827	Yash Contractor	aadi81@example.com
27955158	Abeer Kamdar	aaggarwal@example.net

I created a table visualization that displays customer information including their unique customer ID, full name, and email address. This table serves as a directory, making it easy to access customer contact information for marketing, customer service, or relationship management purposes.

Using the stock intake records, I created a column chart that shows how much stock was received on each day. The horizontal axis represents the days, and the vertical axis shows the total quantity of stock received. This visualization helps monitor inventory inflows and provides insights into supplier performance and stock replenishment cycles.





Percentage of Damaged Stocks

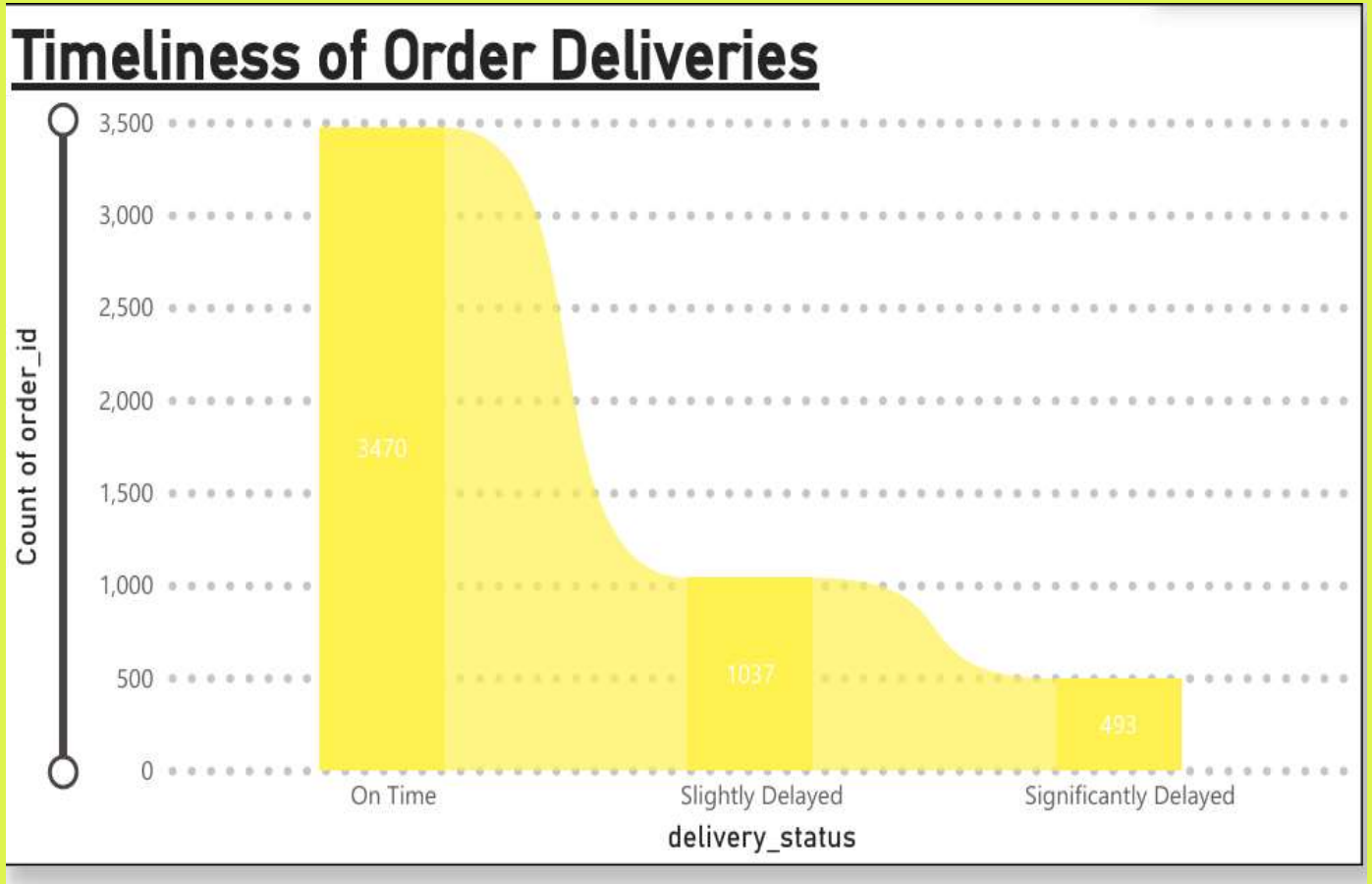
product_name	product_id	Sum of stock_received	Sum o
Vitamins	298985	689	
Cheese	799952	459	
Frozen Biryani	306177	533	
Salt	764014	507	
Toilet Cleaner	336847	444	
Tomatoes	818990	477	
Baby Food	82484	687	
Total		**147526**	

I created a table that displays each product along with the total stock received and the quantity of damaged stock. I then calculated the percentage of damaged stock for each product by dividing the damaged quantity by the total stock received. This table helps identify products with higher damage rates, highlighting quality issues that may need to be addressed with suppliers or handling processes.

I imported campaign data and created a table that displays each marketing campaign along with the total spending and revenue generated from that campaign. This visual allows the business to easily compare the return on investment for different marketing efforts and evaluate which campaigns are most effective in driving sales.

Marketing Campaign Spend vs. Revenue Generated

Campaign name	Sum of spend	Sum of revenue gene
Referral Program	18,18,025.51	36,91
New User Discount	18,33,454.81	36,03
Email Campaign	18,10,729.67	36,01
Total	**1,63,19,838.24**	**3,21,93,**

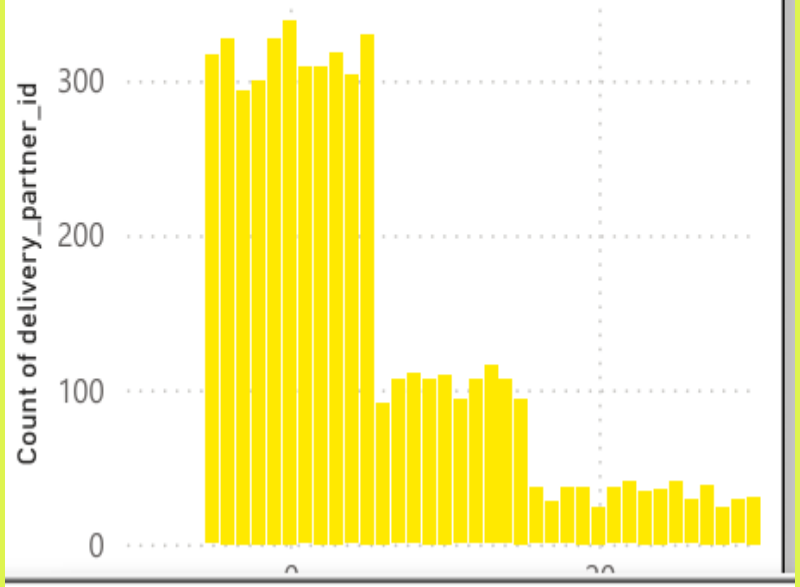


I created a column chart that categorizes orders based on their delivery status. The categories include On-Time Deliveries, Slightly Delayed Deliveries, and Significantly Delayed Deliveries. This chart helps assess the overall efficiency of the delivery process and identify any areas where delays are frequent and need attention

Order Volume by Pincode

Pincode	Order_id(Count)
6749	1
7186	1
13634	1
14359	1
14789	1
Total	5000

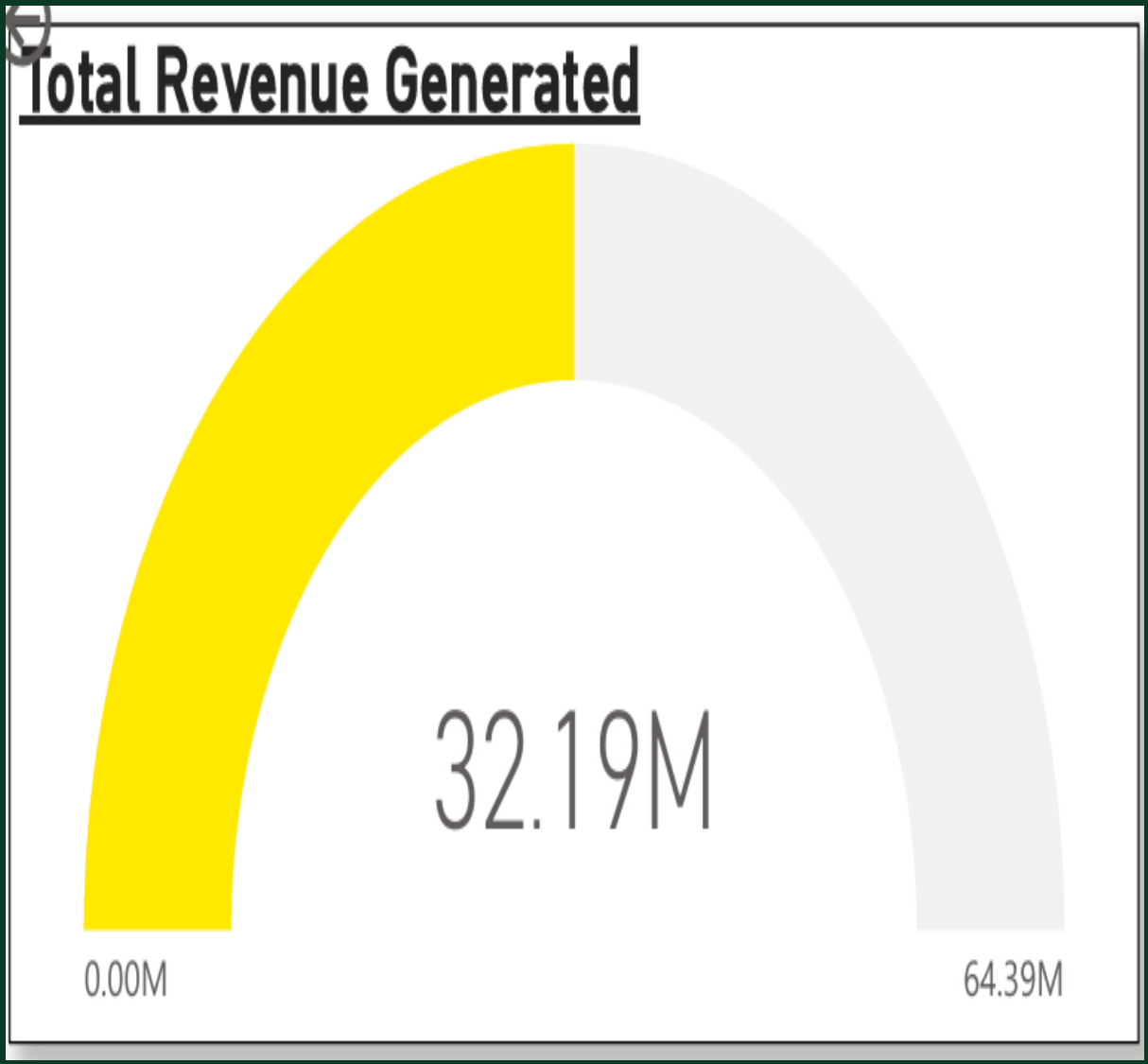
Delivery Delays Distribution

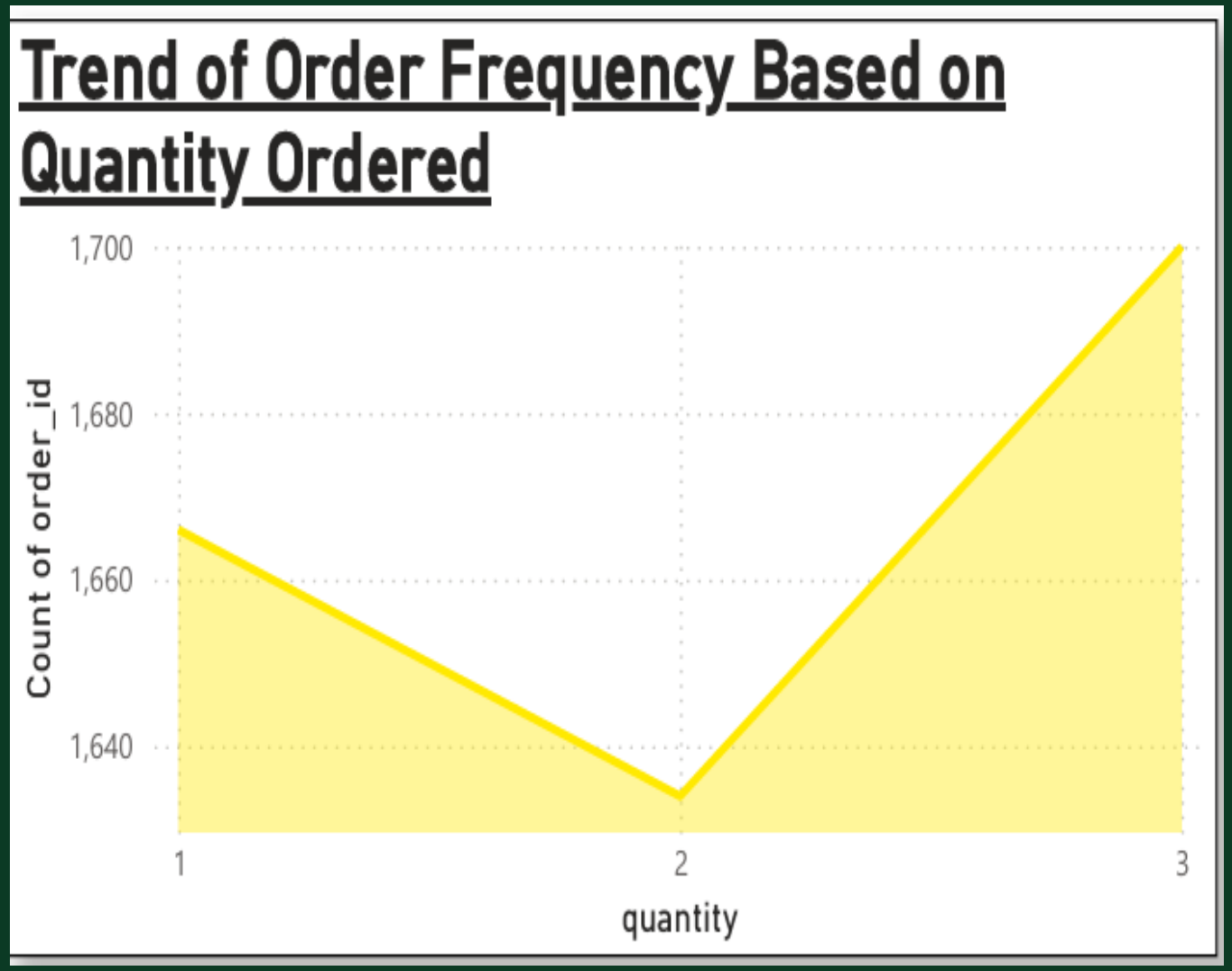


I created a table that lists different delivery areas based on pincodes and shows the total number of orders delivered to each pincode. This information helps analyze geographic demand and identify regions with higher or lower sales volumes, assisting in regional marketing and logistics planning.

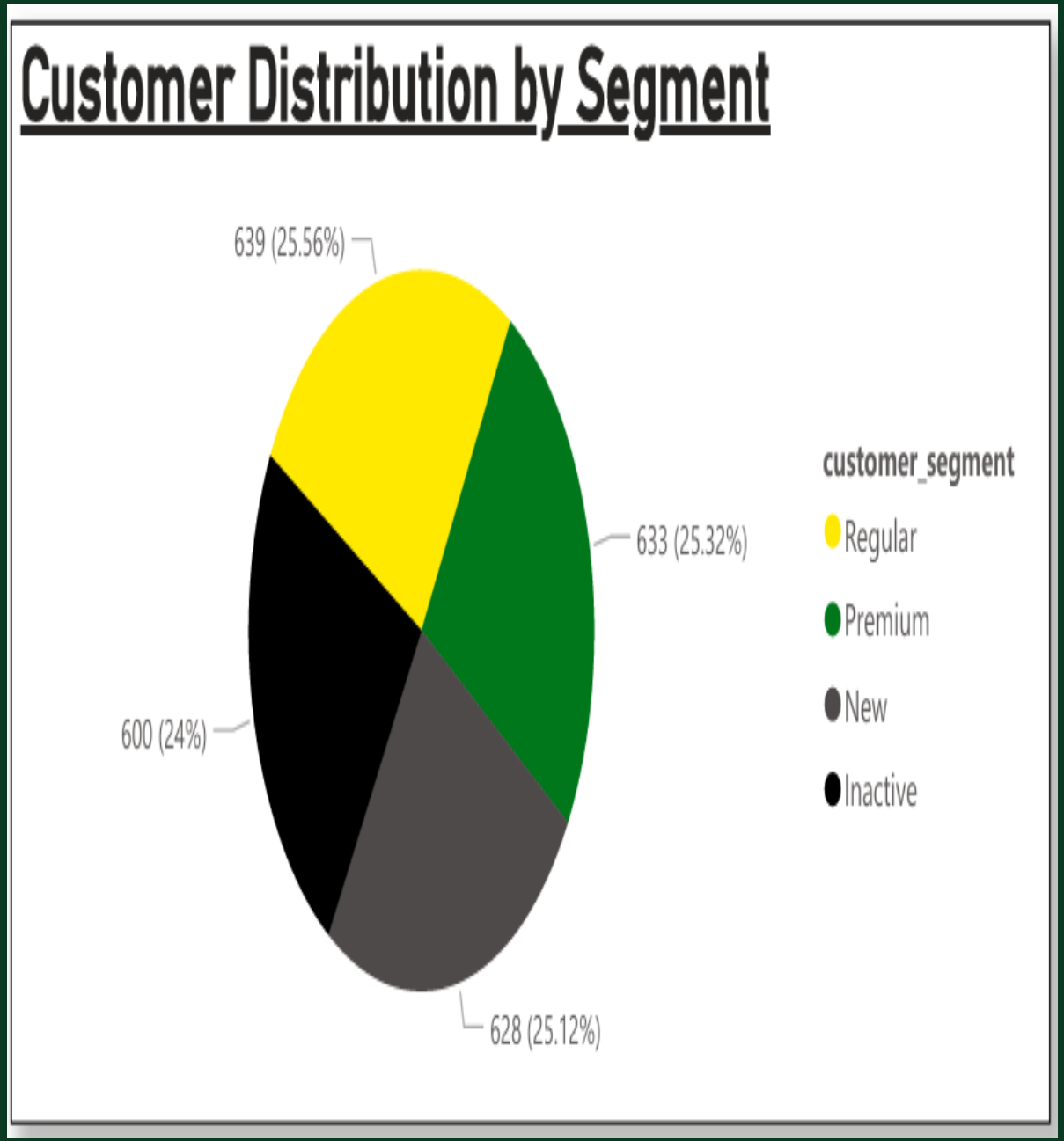
I used delivery time data to create a histogram showing how frequently different levels of delivery delay occurred. The horizontal axis shows the delay time in minutes, while the vertical axis shows how many delivery partners experienced delays of that length. This visualization helps pinpoint common delay ranges and supports analysis for improving delivery performance.

I calculated the total revenue from all sales transactions and displayed it as a KPI visual. This provides a quick summary of the total sales revenue generated, allowing stakeholders to track the financial performance of the business.

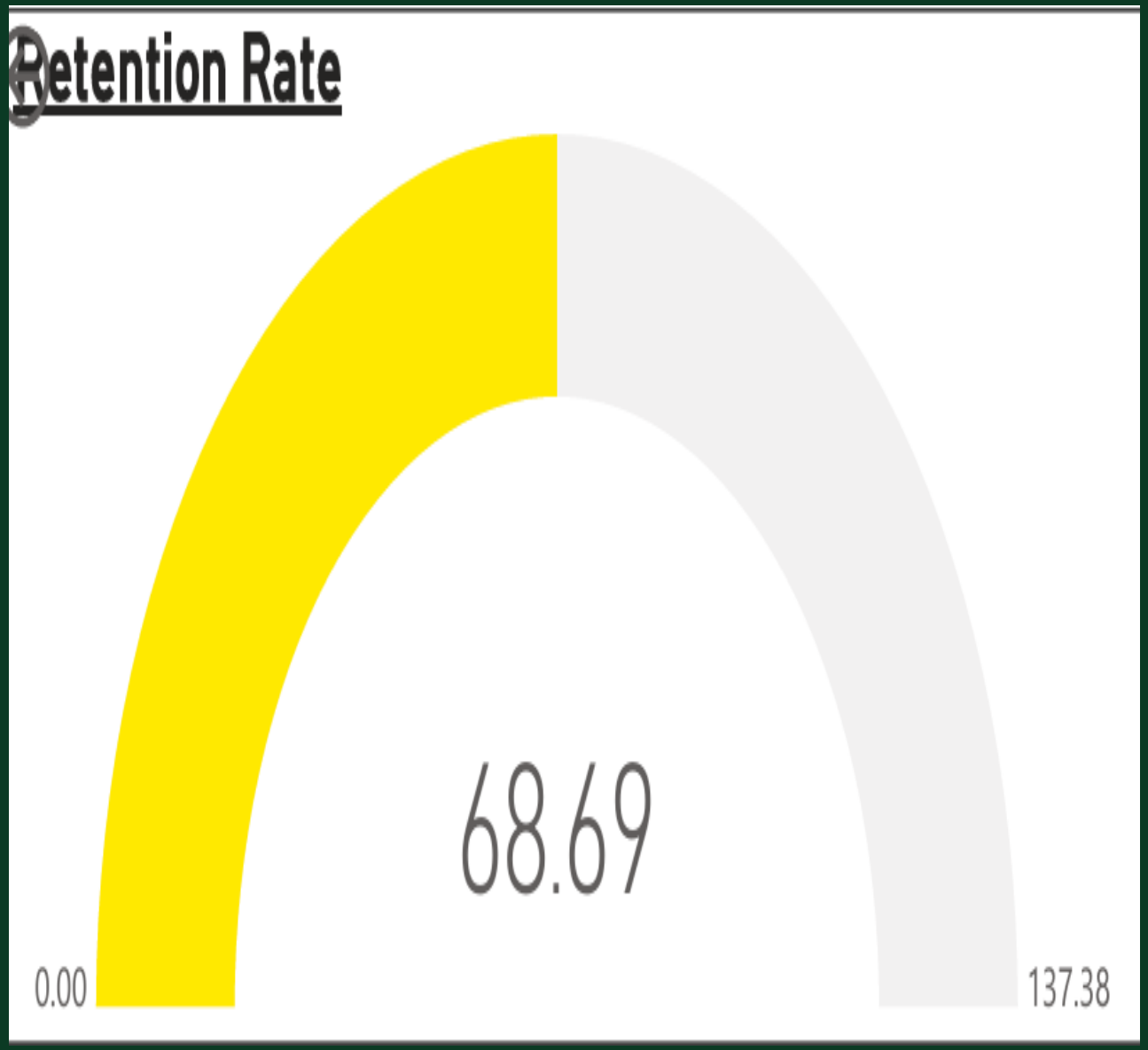




To create this visual, I used the order data and plotted the number of items ordered per transaction against the frequency of such orders. I selected the quantity field for the horizontal axis and counted the total number of orders for each quantity level on the vertical axis. A line chart was used to clearly show the trend. The visualization highlights customer purchasing behavior by indicating which order quantities are most common. This helps in understanding whether customers tend to buy single items or multiple items per order, providing insights that can inform product bundling, inventory planning, and promotional strategies.

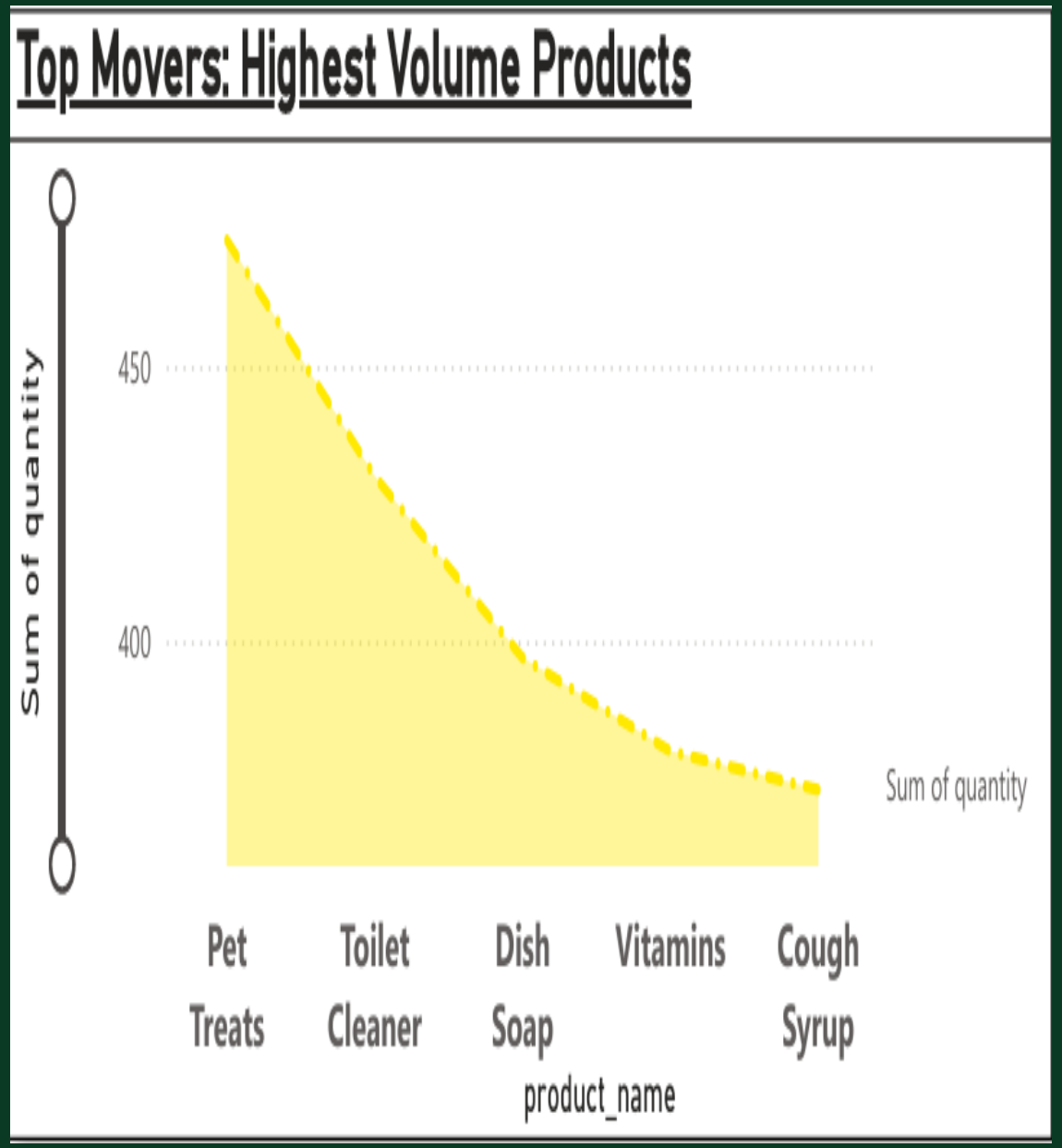


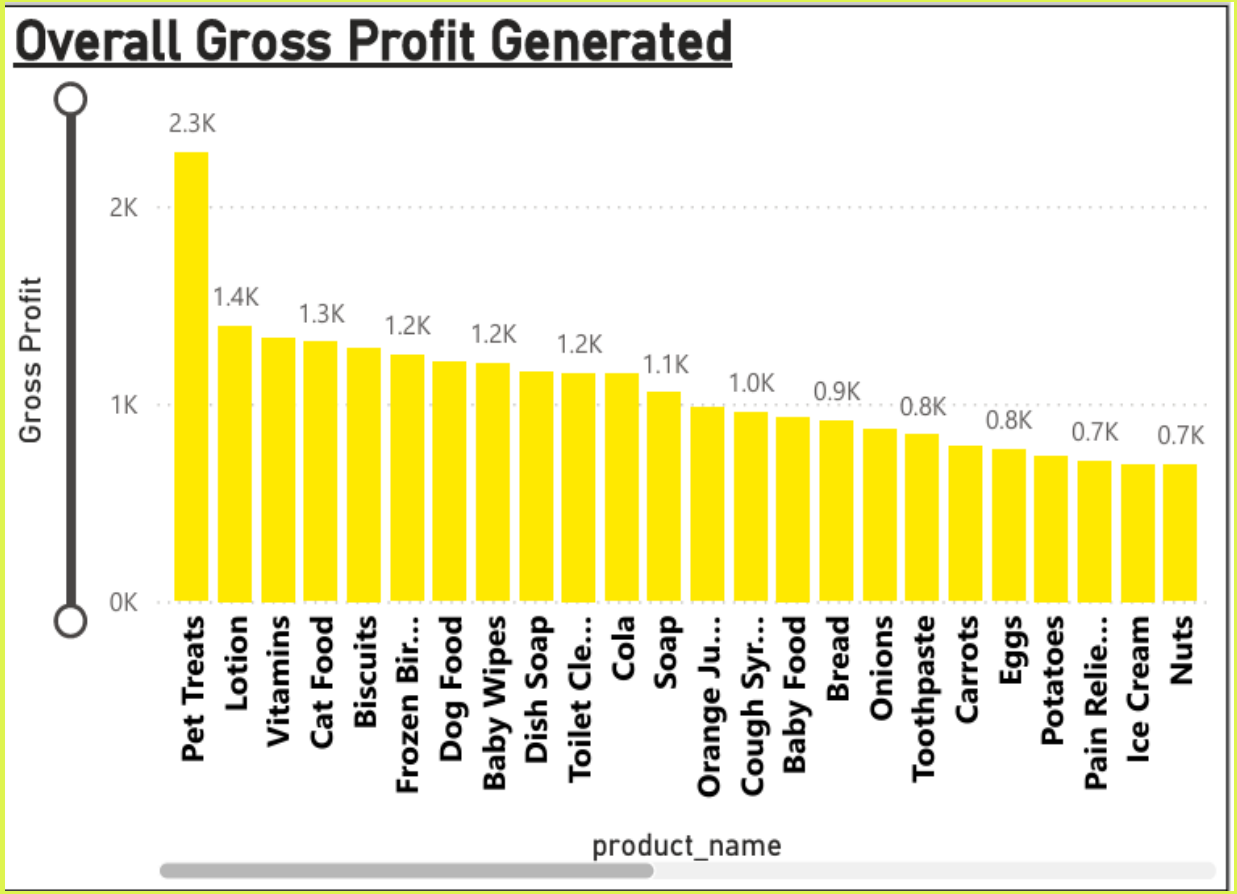
I created a pie chart that shows the proportion of customers belonging to different segments: Regular, Premium, New, and Inactive. This helps visualize the distribution of customer types and supports targeted marketing strategies tailored to each segment.



I calculated the retention rate by analyzing how many customers made repeat purchases over time. The KPI card displays this retention percentage, giving an indication of customer loyalty and satisfaction with the service.

To create this visual, I analyzed product-wise sales data by calculating the total quantity sold for each product. I selected product names for the horizontal axis and the sum of quantities sold for the vertical axis. A line and area chart was used to emphasize the ranking of products based on sales volume. This visualization clearly identifies the top-selling products such as Pet Treats, Toilet Cleaner, Dish Soap, Vitamins, and Cough Syrup. It helps the business recognize which products drive the highest sales volume, enabling better inventory management, promotion targeting, and strategic stocking decisions.





I calculated the gross profit for each product based on its profit margin and total sales, then displayed the results in a bar chart. This visual highlights the most profitable products and helps prioritize which products should receive more focus in marketing and inventory decisions.

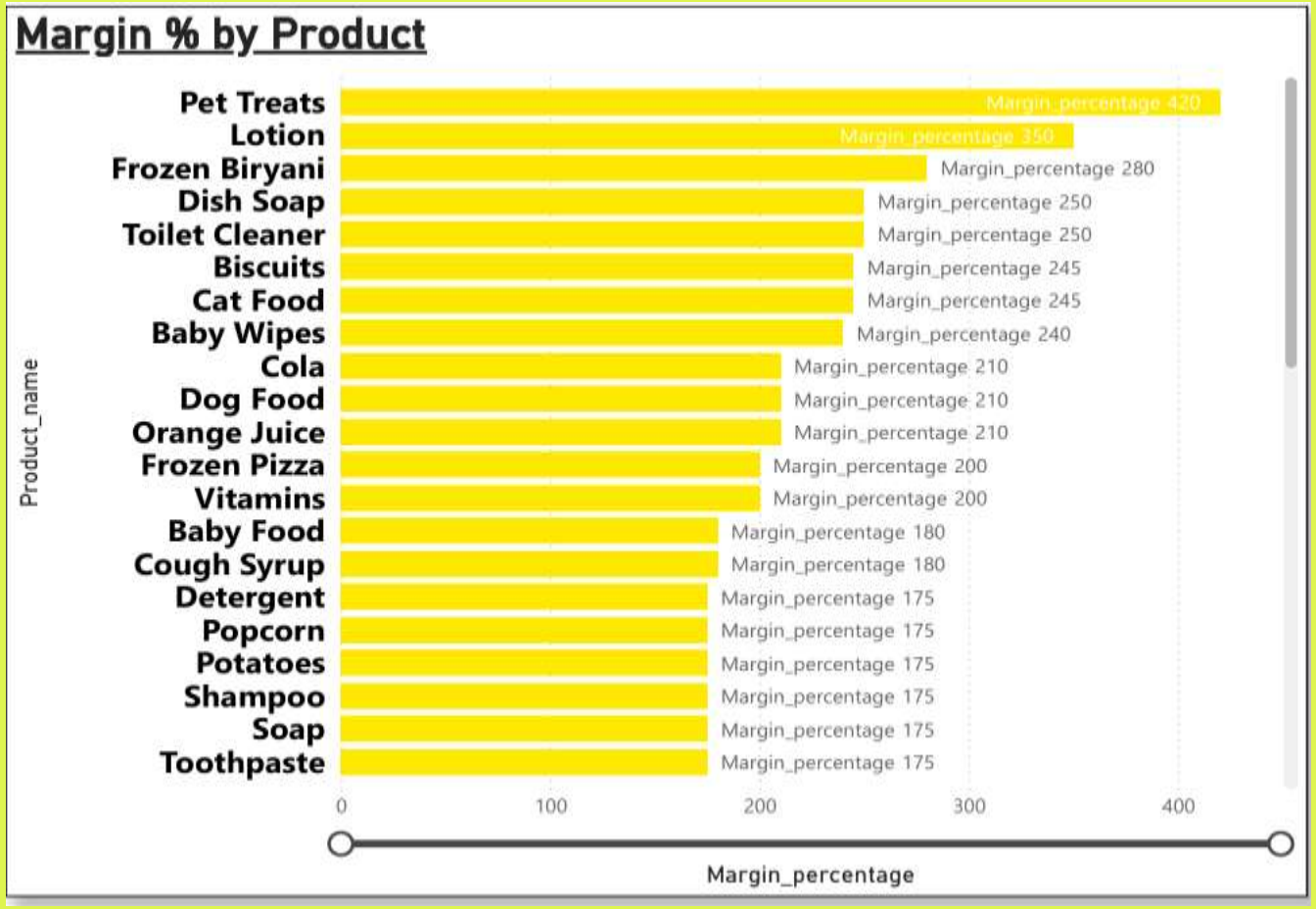
To create this visual, I summarized the total quantity of products sold across different product categories. I used a matrix format where each row represents a category, such as Baby Care, Cold Drinks & Juices, Dairy & Breakfast, Fruits & Vegetables, and Grocery & Staples, while the corresponding column displays the total quantity sold for that category. This visualization provides a clear comparison of sales volumes across various product groups, helping the business identify which categories contribute most to overall sales and prioritize inventory management, marketing efforts, and promotional strategies accordingly.

category	Sum of quantity
Baby Care	655
Cold Drinks & Juices	758
Dairy & Breakfast	1114
Fruits & Vegetables	966
Grocery & Staples	895
Total	10034

To create this visual, I organized sales data to display the total quantity of orders for each brand. The matrix shows a list of brands along with their corresponding total quantities sold. This format enables a detailed comparison of sales performance across different brands within various product categories. By analyzing this visualization, the business can identify which brands are performing well and contributing most to overall sales, providing valuable insights for supplier negotiations, inventory decisions, and targeted marketing initiatives.

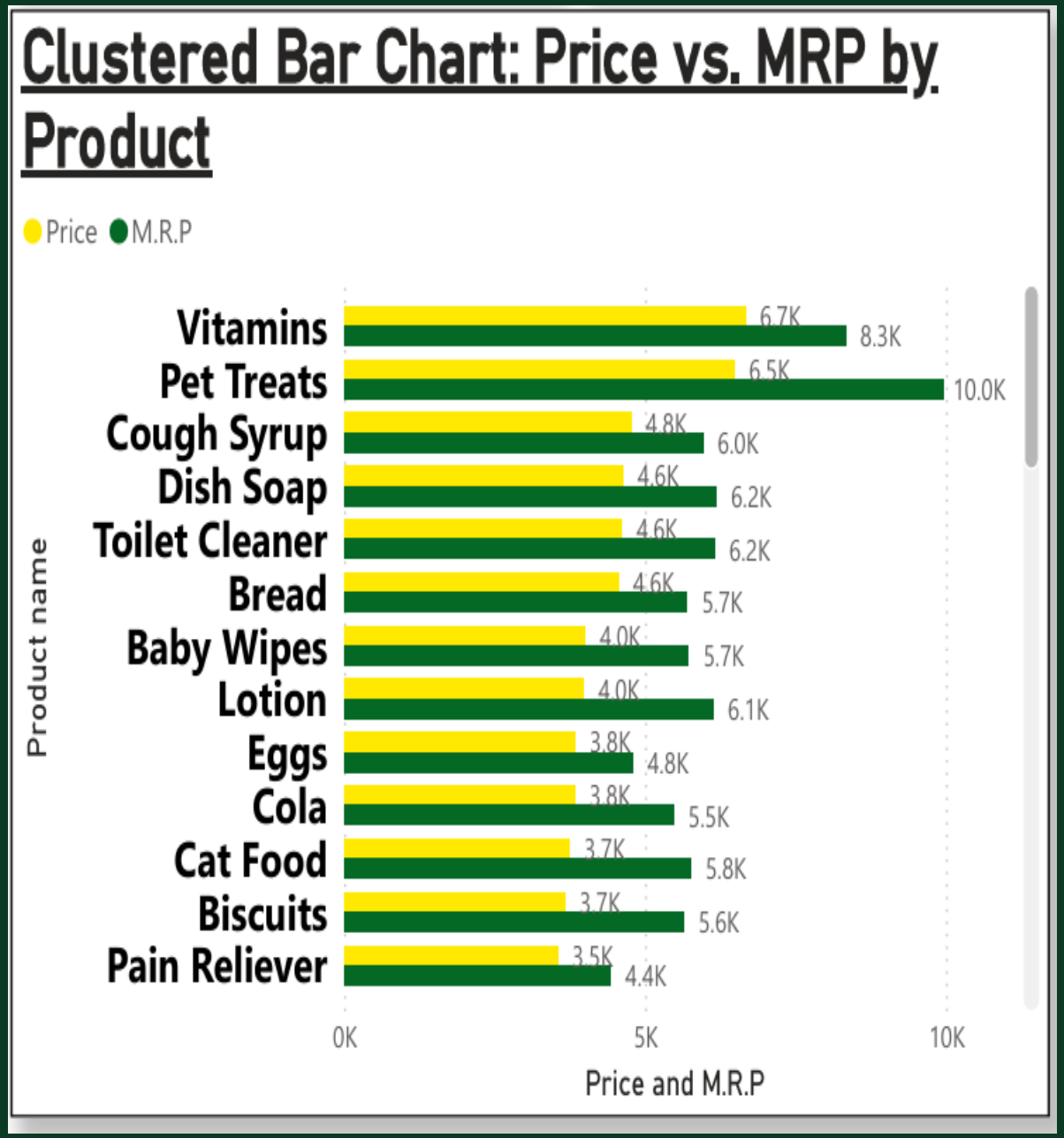
Order Quantity by Product Category and Brand

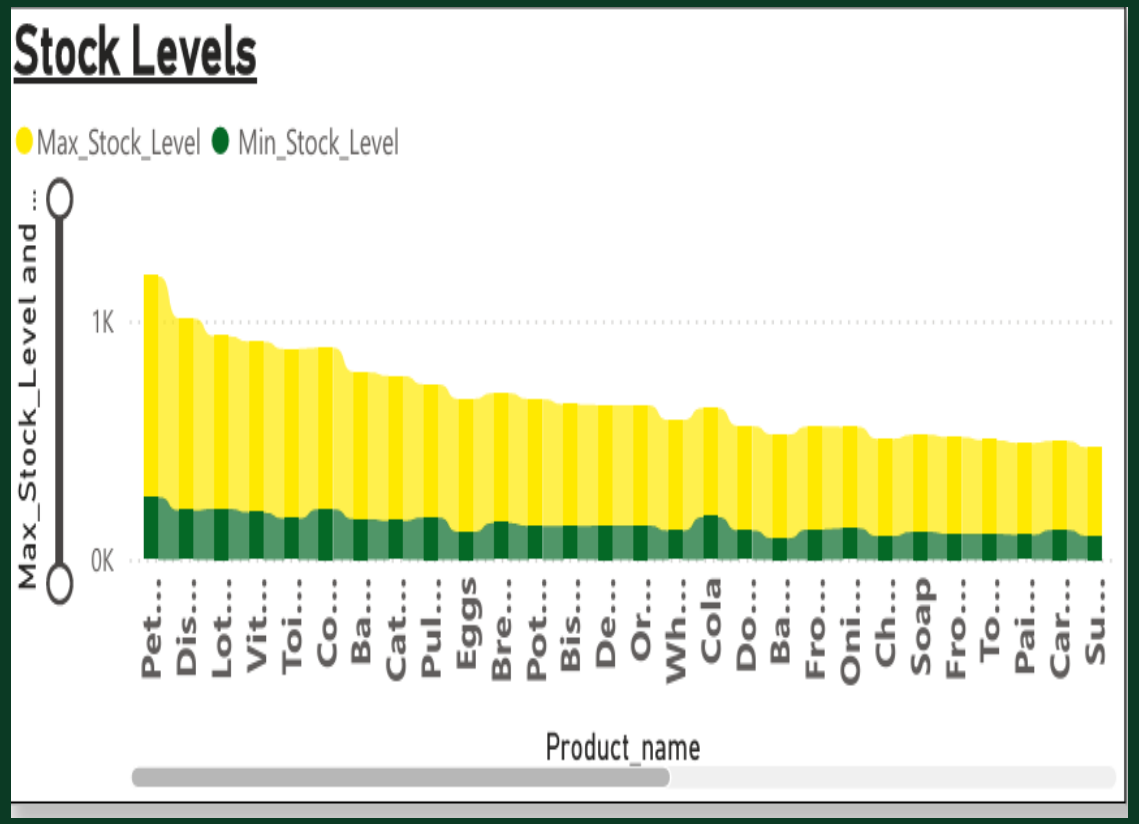
Brand	Quantity
Acharya Ltd	24
Acharya PLC	17
Agarwal-Dhar	19
Agate and Sons	16
Agate Ltd	15
Aggarwal Group	15
Andra, Mander and Kari	13
Anne, Mammen and Khatri	23
Apte, Soman and Raju	16
Arora LLC	17
Arora-Magar	14
Arya Group	14
Arya, Sundaram and Pingle	14
Aurora LLC	19
Total	5000



I created a horizontal bar chart showing the profit margin percentage for each product. This visual makes it easy to compare profit margins across products and identify items with the highest profitability.

I created a clustered bar chart that compares the selling price and maximum retail price (MRP) for each product side-by-side. This visual shows how much discount is offered on each product and helps evaluate pricing competitiveness and promotion strategies.

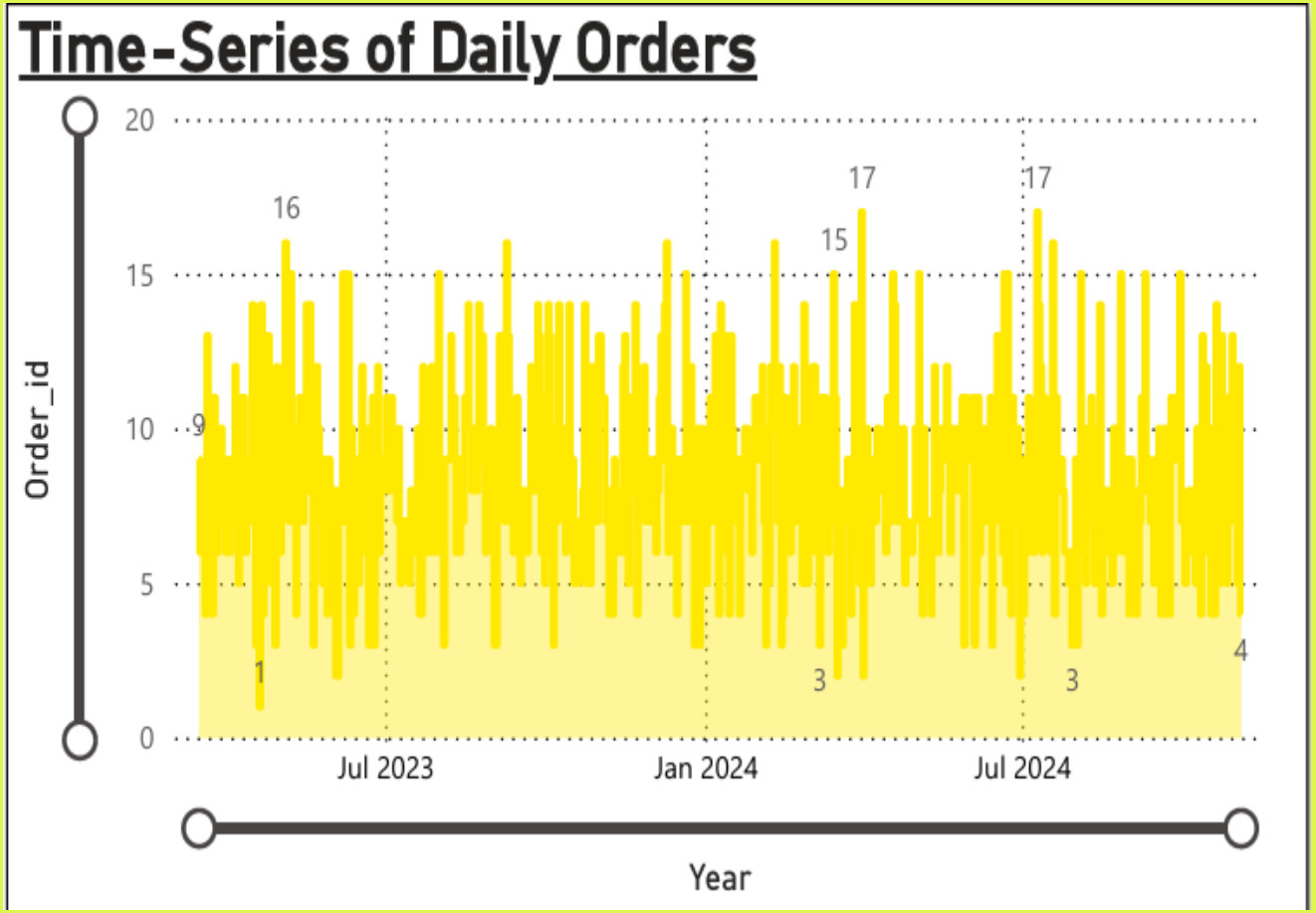




I created a column chart that displays both the maximum stock level and minimum stock level for each product. This helps visualize the inventory threshold for each item and ensures stock levels remain within optimal ranges to avoid stockouts or overstocking.

5000	267	27.78	231.76
Total Products	Total Brands	Average Margin %	Average Shelf Life (Days)

I created four KPI cards to present key summary metrics related to product portfolio and inventory. The first card displays the total number of products available, which is 5000, providing a quick snapshot of the overall product range. The second card shows the total number of distinct brands, which is 267, reflecting the diversity of suppliers contributing to the product assortment. The third card presents the average profit margin across all products, calculated as 27.78%, offering insight into the overall profitability level of the product mix. The fourth card displays the average shelf life of the products, calculated at 231.76 days, which helps in monitoring inventory turnover and managing stock freshness. Collectively, these KPIs provide a high-level overview of product variety, supplier base, profitability, and stock sustainability.

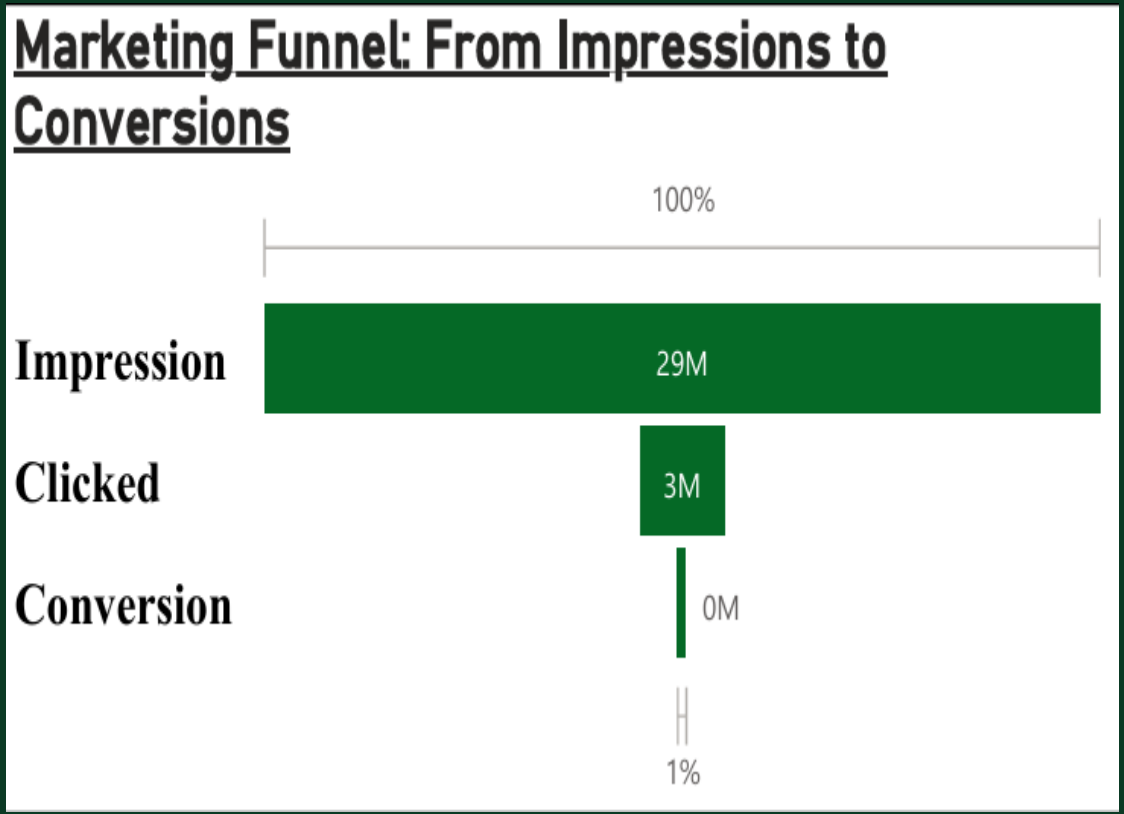


I created this visual by organizing the order data based on the exact order dates and plotting the total number of orders placed each day. A line chart was used to clearly show fluctuations in order volume over time. The horizontal axis represents the calendar dates while the vertical axis shows the number of orders on each day. This time-series visualization helps identify trends, seasonality, and fluctuations in customer purchasing behavior. It allows the business to analyze peak periods of sales, detect any sudden drops or spikes in demand, and supports better planning for inventory management, staffing, and marketing campaigns based on daily order patterns.

I calculated customer lifetime value by multiplying each customer's average order value with the total number of orders they have placed. I presented these results in a table listing the customer name, total orders, average order value, and lifetime value. This helps prioritize high-value customers for loyalty programs and marketing initiatives.

Estimated Customer Lifetime Value

Customer_id	Customer_name	Avg_order_value	Total_orders
45383958	Aachal Mangat	482.19	19
14161586	Aachal Nazareth	353.97	11
15487049	Aadhya Cherian	1,928.00	15
87222820	Aadhya Padmanabhan	497.66	13
65618148	Aadhya Palla	1,644.51	9
35885052	Aadhya Ravi	1,838.87	1
79206969	Aadi Bains	721.13	1
10608845	Aadi Gole	735.36	2
65692224	Aahana Buch	1,053.33	3
Total		**27,55,939.59**	**26229**

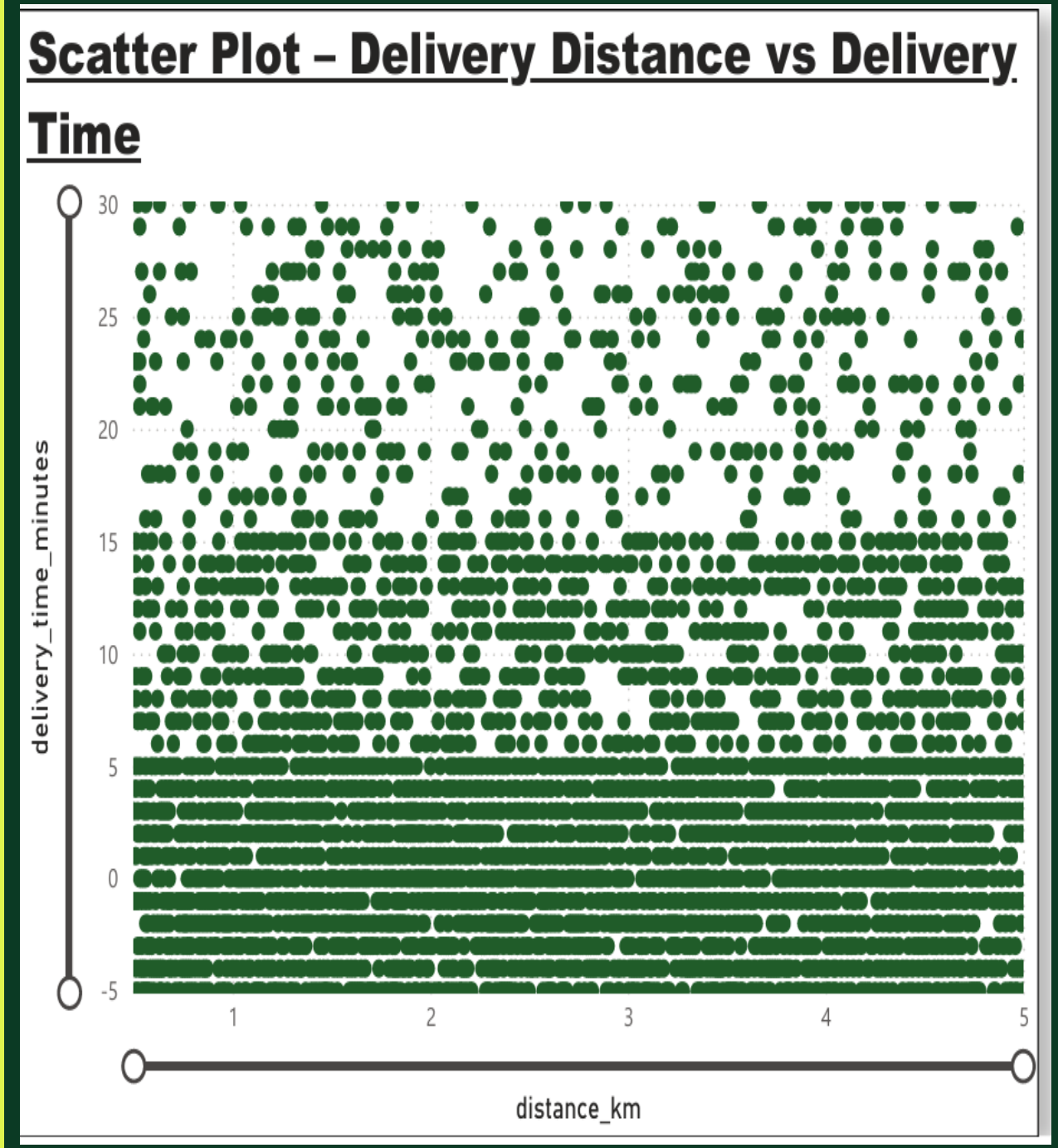


I created a funnel chart that visualizes the customer journey through marketing campaigns. The stages include total impressions, clicks, and conversions. The visual shows how many potential customers progress through each stage, helping to identify drop-off points and optimize the effectiveness of marketing campaigns.

To create this visual, I calculated the average order value by dividing the total revenue generated by the total number of orders placed. I used Power BI's KPI visual to display the calculated average order value as a single metric for quick reference. This visual provides a clear summary of how much revenue, on average, is earned per order. It helps assess overall customer spending behavior and is useful for evaluating pricing strategies, sales performance, and the effectiveness of upselling or cross-selling initiatives. Monitoring this value over time allows the business to track whether customers are increasing their spending per transaction, which directly impacts profitability.

I created a DAX measure to calculate Return on Ad Spend by dividing revenue generated by the marketing spend for each campaign. The result was displayed as a KPI card to track marketing efficiency and investment returns.

I created a scatter plot where each point represents an order, plotted by the distance covered for delivery on one axis and the time taken for delivery on the other. This visual helps analyze the relationship between distance and delivery time, identifying any inefficiencies in the delivery process.



Thank you

Created by-
Shreya Das.